

FedMultiMed: A Federated Multi-Modal Ensemble Framework for Medical Image Classification with Explainable AI

Presented by

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Outline

- Introduction
- Objective
- Background Study
- Gap Analysis
- Methodology
- Sample Dataset
- Results & Analysis
- Novelty of the Work
- Conclusion
- References

Introduction

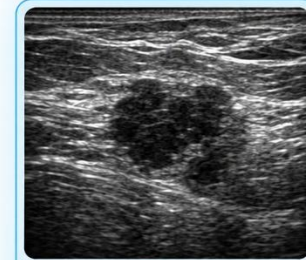
Medical imaging plays an important role in early disease diagnosis and modern healthcare. Deep learning has achieved strong results in medical image classification, but many existing systems focus on only one imaging modality and often depend on centralized data storage, which can create privacy and data-sharing concerns. Different medical images, such as brain MRI, chest X-ray, breast ultrasound, and dermoscopy, also have different visual characteristics and class distributions. To address these challenges, FedMultiMed combines federated learning, ensemble model, and Explainable AI in a single multi-modal medical image classification framework.



Brain MRI



Chest X-ray



Breast Ultrasound



Dermoscopy



Four Modalities, One Privacy-Preserving Framework

Raw medical images remain at local clients.

Objective

OBJECTIVE 1 — Multimodal Evaluation & Complementary Backbone Selection

To evaluate pretrained CNN models and select optimal backbones for multimodal medical image classification across brain MRI, chest X-ray, breast-ultrasound, and dermoscopic images.

OBJECTIVE 2 — Privacy-Preserving Federated Ensembling Framework

To develop a privacy-preserving federated learning framework using the selected CNN backbones that evaluates multiple aggregation methods and builds a calibrated hybrid ensemble without transferring raw patient data.

OBJECTIVE 3 — Diagnostic Validation & Quantitative Explainable AI

To evaluate the diagnostic performance through held-out testing and cross-validation, and analyze prediction decisions using Grad-CAM, Grad-CAM++, Integrated Gradients, and LIME.

Background Study

Author(s)	Year	Title	Methodology	Key Findings
Haripriya et al. [1]	2025	Privacy-Preserving Federated Learning for Collaborative Medical Data Mining	EfficientNetV2 and ResNet-RS were trained using adaptive FedAvg and FedSGD based on client-data divergence.	Achieved up to 96.3% accuracy, but used fixed clients and communication rounds.
Muntaqim and Smrity [2]	2025	Federated Learning Framework for Brain Tumor Detection Using MRI Images	VGG19 was trained across four non-IID clients using FedAvg, FedProx, and SCAFFOLD with Grad-CAM.	SCAFFOLD achieved 98.45% accuracy, but the study covered only brain MRI with simulated clients.
Borazjani et al. [3]	2025	Multi-Modal Federated Learning for Cancer Staging Over Non-IID Datasets	Distributed gradient blending and proximity-aware weighting combined histopathology, clinical, and molecular data.	Improved accuracy by about 10–15 percentage points, but validation was limited to TCGA cohorts.
Fan et al. [4]	2025	A Personalized Multimodal Federated Learning Framework for Skin Cancer Diagnosis	Hierarchical aggregation, modality imputation, and attention-based fusion were used for incomplete multimodal data.	Achieved 92.32% accuracy and reduced communication by 32.9%, but focused only on skin cancer.
Gupta et al. [5]	2025	Applying YOLOv6 as an Ensemble Federated Learning Framework	Pruned YOLOv6 models were trained across five clients with ensemble prediction and homomorphic encryption.	Achieved 98% and 97% validation accuracy, but covered only breast-related datasets.
Pervez et al. [6]	2025	Towards Trustworthy AI-Driven Leukemia Diagnosis	EfficientNetB3 was trained through hierarchical federated learning and evaluated with 18 XAI methods.	Quantitatively evaluated explanation quality, but considered only microscopic blood-cell images.

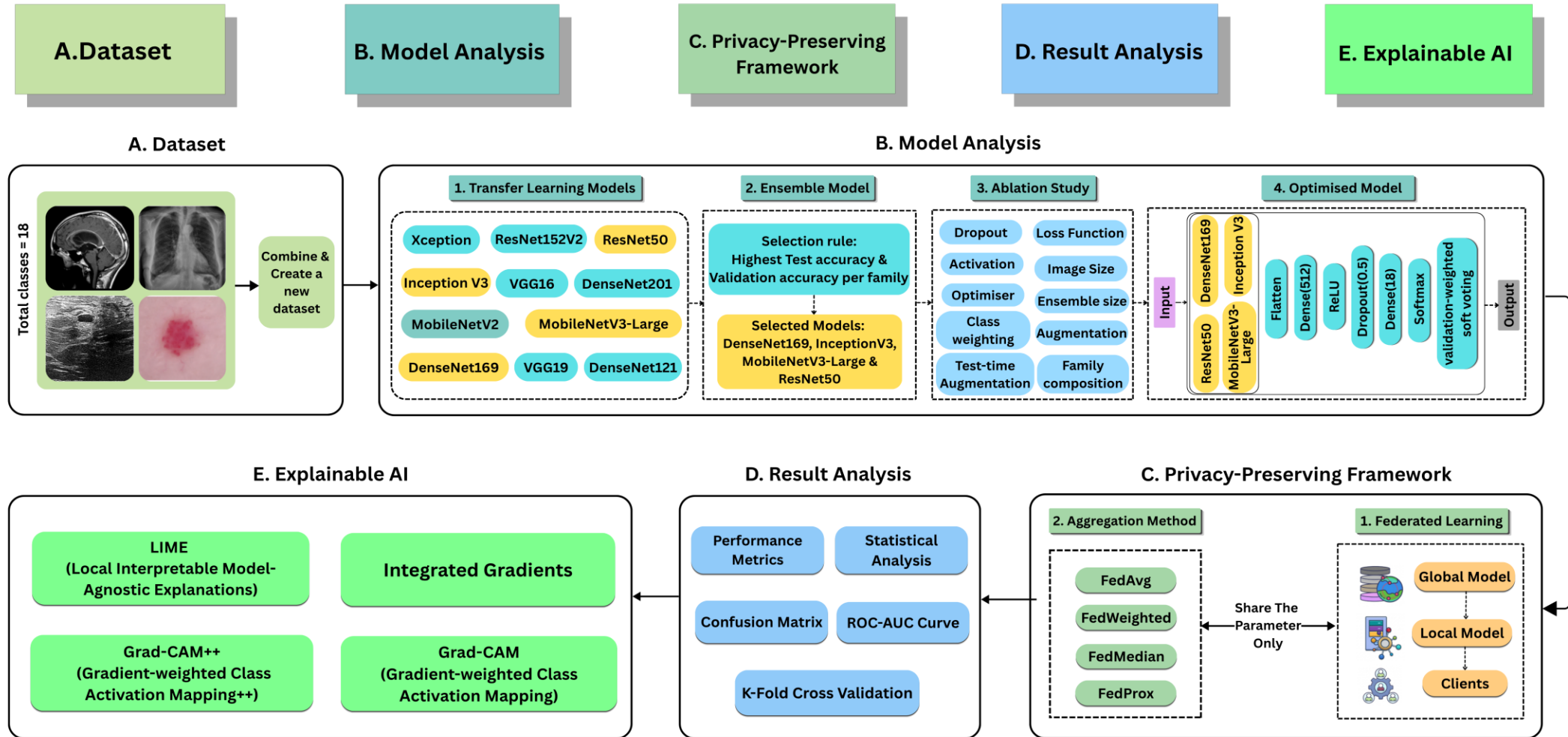
Gap Analysis

Study	Multi-modal	Federated	Ensemble	XAI	Aggregation Compared	Leakage-Controlled
Haripriya et al. [1]	Yes	Yes	No	No	Yes	No
Muntaqim & Smrity [2]	No	Yes	No	Yes	Yes	No
Borazjani et al. [3]	Yes	Yes	No	No	No	No
Fan et al. [4]	Yes	Yes	No	No	Yes	No
Gupta et al. [5]	Yes	Yes	Yes	No	No	No
Pervez et al. [6]	No	Yes	No	Yes	No	No
Proposed FedMultiMed	Yes	Yes	Yes	Yes	Yes	Yes

KEY TAKEAWAY FROM GAP ANALYSIS:

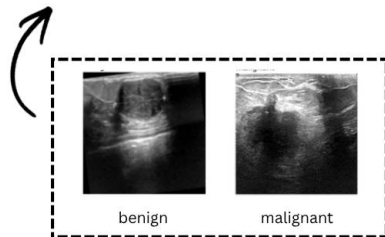
The reviewed studies address individual areas such as multimodality, federated Learning, ensemble models, or explainable AI. However, none of them combines all these features together in a single framework. FedMultiMed addresses this integration gap within one framework.

Methodology

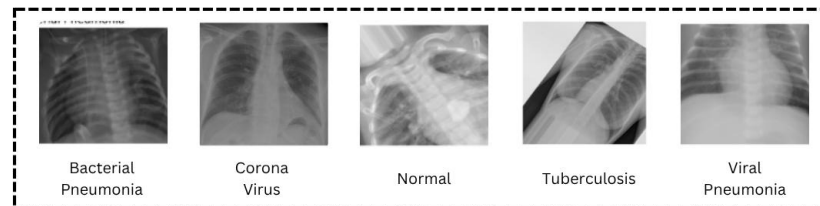
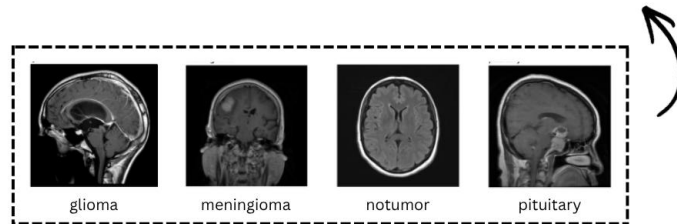


Sample dataset

Untrasound Breast Cancer

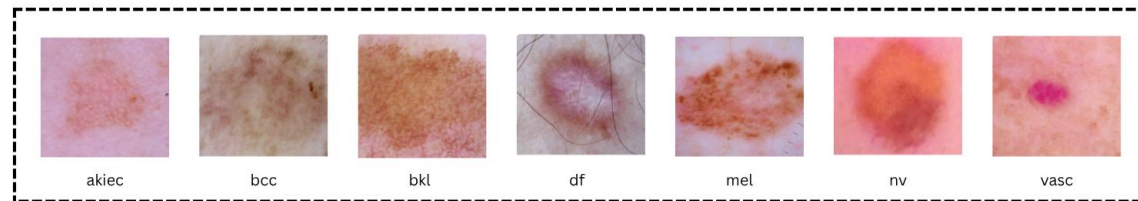


Brain Tumor MRI



Lung Disease X-ray

Skin Cancer Dermoscopy

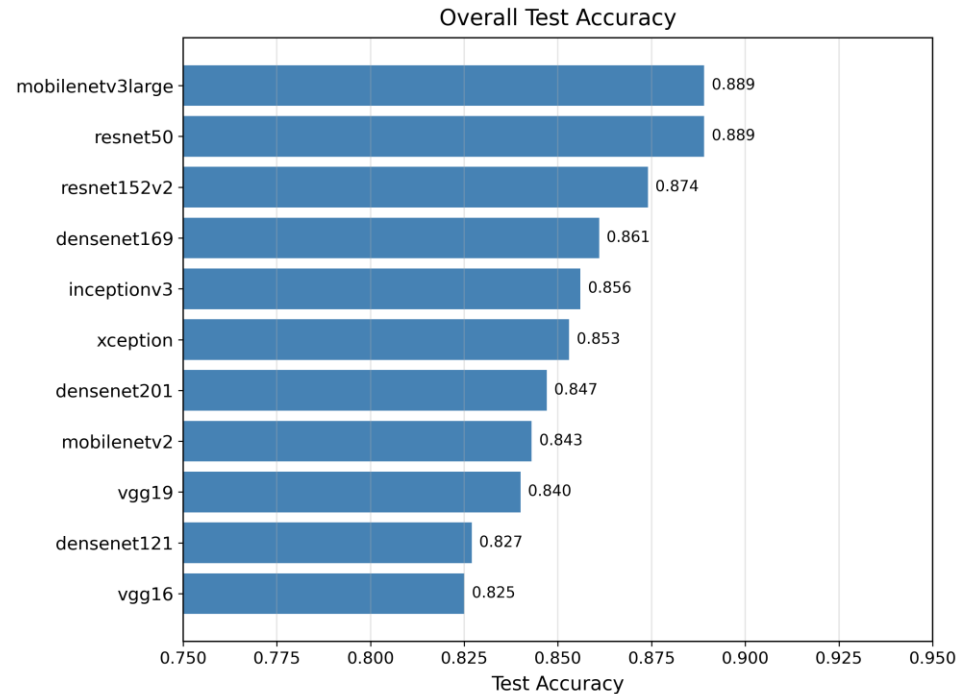


Modality	Classes	Images
Brain MRI	4	7,200
Chest X-ray	5	10,095
Breast Ultrasound	2	9,016
Dermoscopy	7	10,015
Total	18	36,326

36,326 images across 4 modalities

Result and Analysis

Candidate Backbone Benchmark (11 Models)



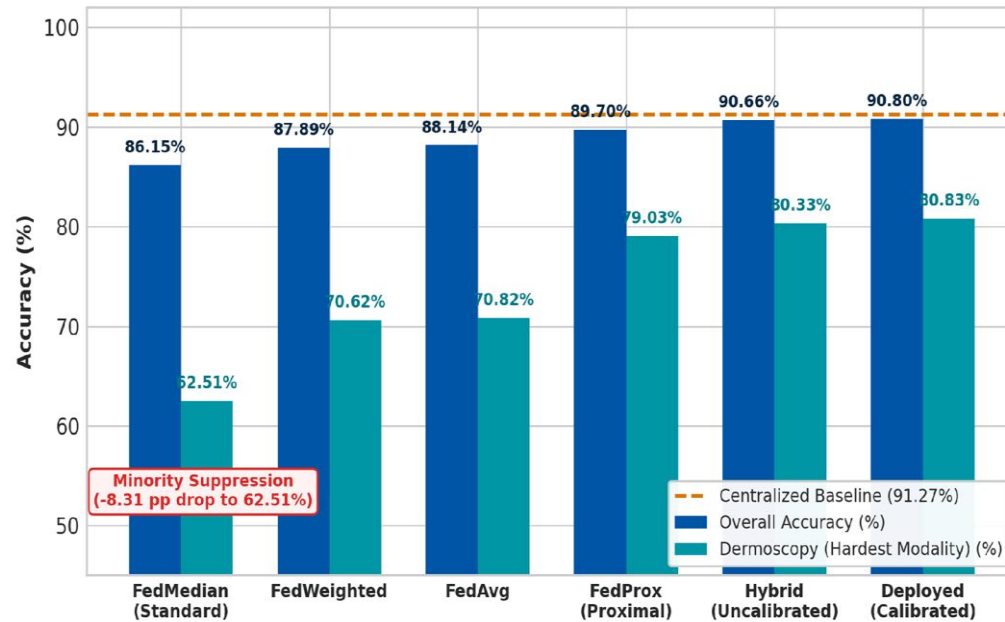
Class Weighting and Loss-Function Ablation

Configuration	Accuracy	Macro-F1	Dermoscopy
Square-root balanced weighting	0.9127	0.8100	0.8123
Label smoothing	0.9068	0.8310	0.7903
No weighting	0.9017	0.7620	0.7903
Balanced weighting	0.8941	0.8227	0.7523
Focal loss	0.4722	0.1694	0.6697
Kullback–Leibler divergence	0.4711	0.1766	0.6692

- **Square-Root Weighting Selected:** Achieved top accuracy (0.9127) and top dermoscopy performance (0.8123), balancing severe lesion imbalance.
- Catastrophic Loss Collapse: Focal loss and KL divergence completely collapsed on the 18-class corpus (accuracy dropped to ~0.47; MRI collapsed to 0.25 chance).
- F1 vs. Accuracy Trade-Off: Label smoothing yielded higher macro-F1 (0.8310) but reduced overall accuracy and dermoscopy (0.7903), favoring square-root weighting.

Result and Analysis

Federated Aggregation Progression & Modality Recovery



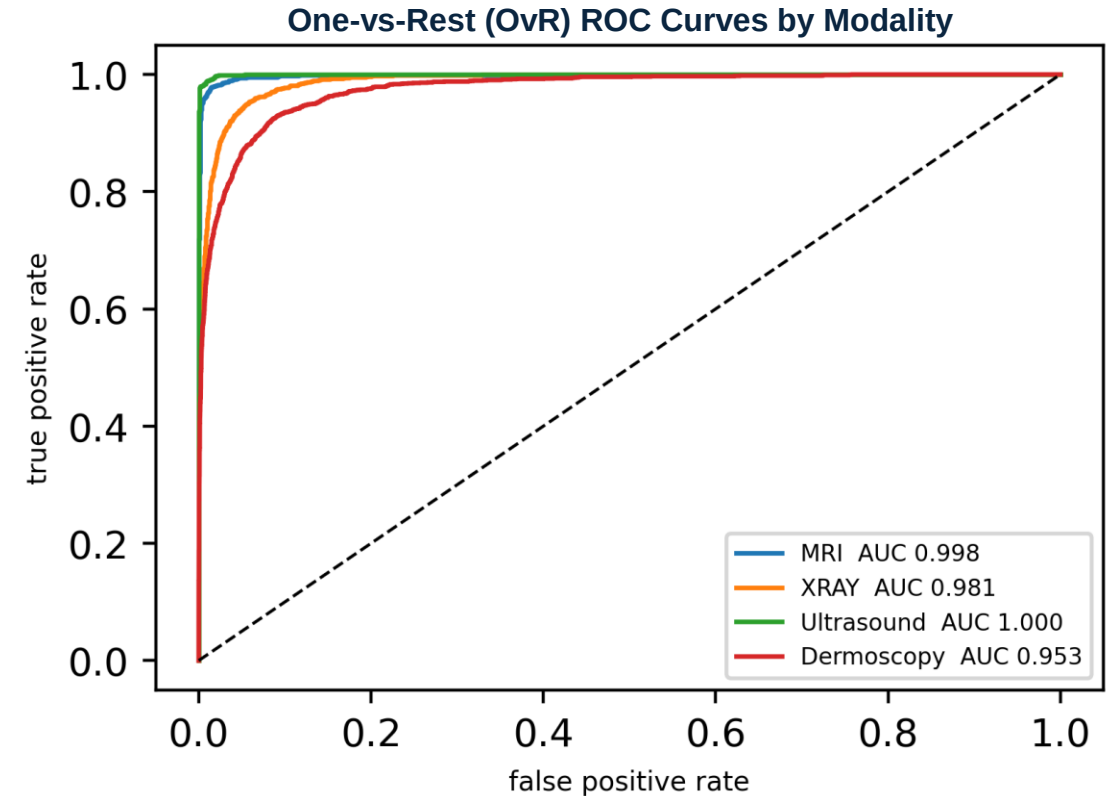
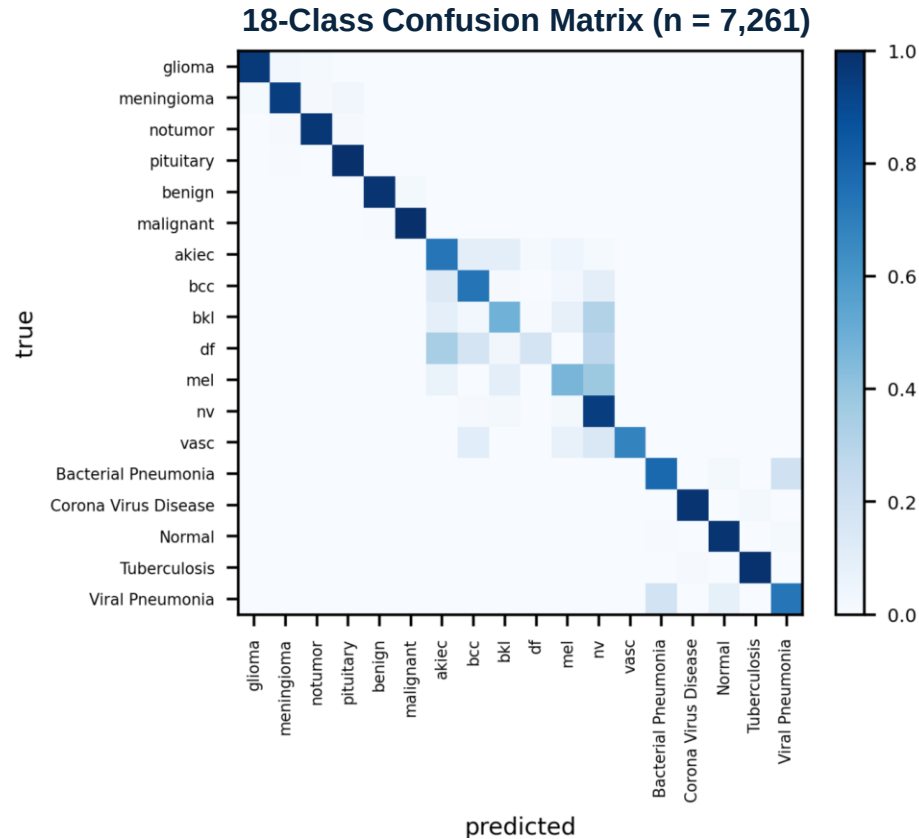
Aggregation Strategies vs. Centralized Baseline (91.27%)

Strategy	Overall Acc	Dermoscopy	vs Centralized
Centralized Baseline	91.27%	81.23%	0.00 pp
FedMedian	86.15%	62.51%	-5.12 pp (Failed on skin)
FedWeighted	87.89%	70.62%	-3.38 pp
FedAvg	88.14%	70.82%	-3.13 pp
FedProx	89.70%	79.03%	-1.57 pp
Hybrid (Uncalibrated)	90.66%	80.33%	-0.61 pp
Hybrid (Deployed Calibrated)	90.80%	80.83%	-0.47 pp (Final Model)

CORE RESEARCH DISCOVERY & PRIVACY-PERFORMANCE BALANCE:

Standard FedMedian severely penalizes minority skin cancer (-8.31 pp drop to 62.51%). Our proposed Hybrid FedProx + Temperature Calibration successfully recovers Dermoscopy to 80.83% and delivers 90.80% overall accuracy, maintaining raw-data privacy with only a 0.61 pp federation cost against the 91.27% centralized baseline.

Result and Analysis



- **Near-Zero Cross-Modality Error: Only 1 of 7,261 test predictions crossed modalities (0.014% error rate).**
- **High Diagnostic Discrimination: Overall Macro-AUC of 0.9940 (Ultrasound 1.000, MRI 0.998, CXR 0.981, Dermoscopy 0.953).**
- **Clinically Confined Errors: Residual misclassifications strictly occurred within adjacent disease subtypes.**

Result and Analysis

90.80%

TEST ACCURACY

[90.13%, 91.43%] 95% CI

0.8136

MACRO-F1 SCORE

Balanced across 18 classes

0.9940

MACRO-AUC (OvR)

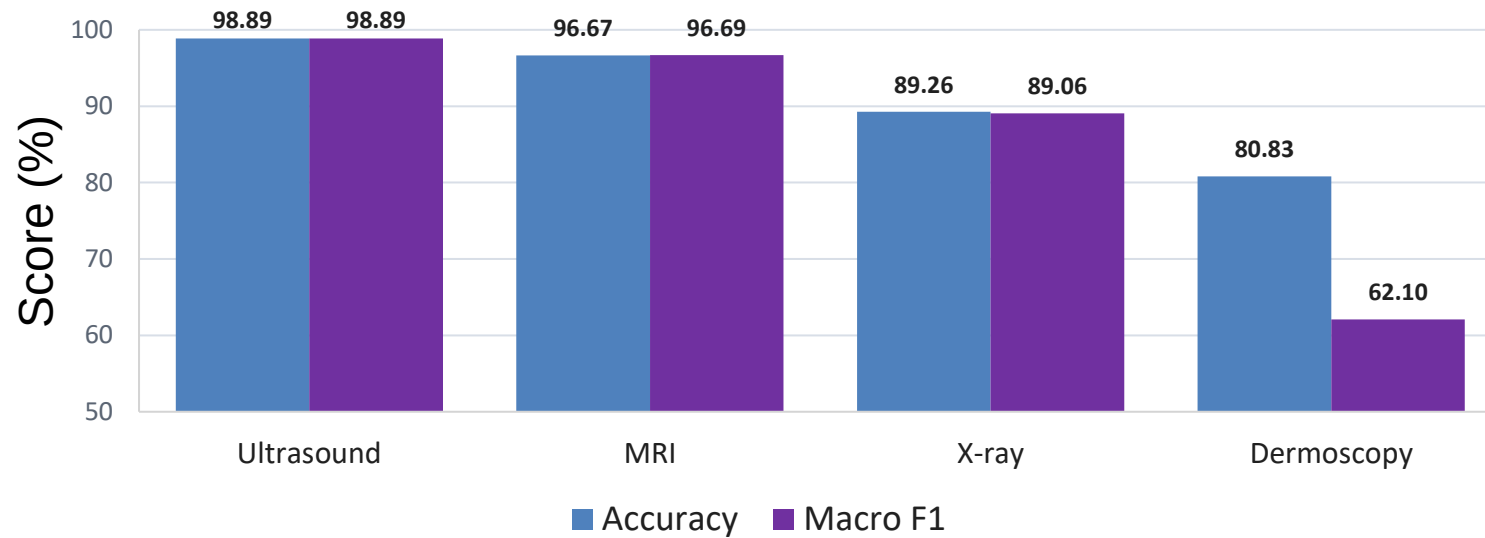
Class discrimination

91.02%

5-FOLD CV ACCURACY

± 0.52% across partitions

Performance Breakdown by Modality (7,261 Test Images)



Novelty of the Work

Federated Multi-Modal Learning Framework

FedMultiMed combines brain MRI, chest X-ray, breast ultrasound, and dermoscopy within one 18-class federated learning framework using modality-partitioned non-IID clients.

Privacy-Conscious and Leakage-Controlled Design

Raw training images remain within their local clients, while only model parameters are shared. Lesion-grouped partitioning also prevents HAM10000 samples from leaking across data splits.

Architecture-Aware Hybrid Ensemble

Four aggregation methods are evaluated across different CNN backbones. The strongest strategy for each backbone is used to construct a calibrated soft-voting ensemble.

Quantitatively Evaluated Explainability

Grad-CAM, Grad-CAM++, Integrated Gradients, and LIME are evaluated through faithfulness, localisation, and sensitivity analysis, providing evidence beyond visual heatmaps.

Conclusion

FedMultiMed demonstrates that federated learning, complementary CNN backbones, and explainable AI can be integrated into one multi-modal medical-image classification framework without transferring raw training images. The calibrated hybrid ensemble achieved 90.80% accuracy, 0.8136 macro F1, and 0.9940 macro AUC on 7,261 held-out images, while five-fold subject-grouped validation reached $91.02 \pm 0.52\%$ accuracy. The results are strong across MRI, X-ray, and ultrasound, although minority dermoscopy classes remain difficult. Therefore, the framework is suitable as a research decision-support system, but further external clinical validation and stronger privacy mechanisms are needed before real-world deployment.



References

- [1] R. Haripriya, N. Khare, and M. Pandey, “Privacy-preserving federated learning for collaborative medical data mining in multi-institutional settings,” *Scientific Reports*, vol. 15, no. 1, Apr. 2025, doi: 10.1038/s41598-025-97565-4.
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- [4] S. Fan, A. Ahmed, X. Zeng, R. Xi, and M. Hou, “A Personalized Multimodal Federated Learning Framework for Skin Cancer Diagnosis,” *Electronics*, vol. 14, no. 14, pp. 2880–2880, July 2025, doi: 10.3390/electronics14142880.
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THANK YOU